QUALCOMM® SNAPDRAGON® AI LAB CHALLENGE 2026

NETRA

Offline AI Vision Assistant for Snapdragon PCs

An accessible, 100% on-device perception engine converting live camera feeds into real-time spoken descriptions via YOLOv8 object detection, EasyOCR, and native Windows SAPI TTS.

Target Hardware: Qualcomm® Snapdragon® Copilot+ PCs | Submission Package

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The Problem: Visual Information Friction in Daily Life

01. Reading Printed Text

Over 2.2 billion people live with vision impairment. Identifying medication dosages, food labels, warning signs, and printed documents creates daily accessibility barriers.

02. Cloud & Network Latency

Existing visual assistant apps depend heavily on cloud LLM APIs, causing high latency, recurring subscription costs, and complete failure during network outages.

03. Privacy & UI Complexity

Streaming live camera feeds to third-party cloud servers poses severe privacy risks. Complex touch interfaces are unusable for individuals requiring instant audio feedback.

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The Solution: Netra Offline AI Perception Engine

100% Offline Vision Assistance

- Zero Cloud Dependency: All processing runs locally on the host PC.
- Zero Subscription Fees: Open-source local perception pipeline.
- Privacy-First Telemetry: Camera frames never leave local memory.
- Instant Audio Feedback: Designed for single-key hands-free operation.

Multi-Modal Local Fusion

- Object Recognition: YOLOv8-N ONNX detects physical items & bounding boxes.
- Printed Text OCR: EasyOCR extracts labels, signs, and document text.
- Local Composition: Rule-based synthesis avoids LLM hallucination.
- Native Speech Output: Windows SAPI speaks clear audio descriptions.

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How Netra Works: 5-Stage Local AI Pipeline

1. Camera Capture

Webcam Video Input (DirectShow / OpenCV)

2. Object Detection

YOLOv8-N ONNX (Bounding boxes & labels)

3. Text Recognition

EasyOCR Engine (Printed text extraction)

4. Local Synthesis

Response Composer (Natural sentence merger)

5. Spoken Output

pyttsx3 / SAPI (Native Windows TTS)

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Technical Architecture & Component Stack

Core Application: Python 3.10 / OpenCV 5.0

Handles camera feed, bounding box overlay, keybindings, and high-contrast dashboard UI.

Inference Engine: ONNX Runtime 1.23

Pre-configured for Qualcomm QNN Execution Provider (Hexagon NPU) with CPU fallback.

Vision Perception: YOLOv8-N ONNX (12.26 MB)

Compact 80-class object detector running letterbox preprocessing and NMS postprocessing.

OCR & Speech: EasyOCR + pyttsx3 / SAPI

Fully offline text extraction and native Windows speech synthesis without cloud latency.

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Accessibility Focus & Real-World Use Cases

Medication & Label Reading

Point camera at prescription bottles or packages; Netra identifies the bottle and reads dosage text out loud.

Object & Obstacle Identification

Recognizes physical objects (chairs, cups, laptops, keys) in immediate surroundings to aid navigation.

Signage & Directional Guidance

Extracts room numbers, office door signs, and printed directions for low-vision individuals.

Zero-Telemetry Privacy

Operates entirely offline, guaranteeing personal camera feeds remain strictly private on-device.

Qualcomm® Snapdragon® Hardware Strategy & Verified Status

Qualcomm QNN EP Strategy

- Qualcomm QNN Integration: Engine requests QNNExecutionProvider (QnnHtp.dll burst mode) targeting Hexagon NPU.
- Snapdragon Copilot+ Target: Designed for ARM64 Snapdragon Windows PCs with onnxruntime-qnn installed.
- Graceful Fallback: Seamlessly defaults to CPUExecutionProvider on non-Snapdragon host development machines.

Empirical Verified Metrics

- Detection Latency: ~40.1 ms average inference time (~25 FPS) on host processor.
- Model Footprint: Compact YOLOv8-N ONNX binary requires only 12.26 MB storage.
- Zero API Dependency: Entire visual perception turn completes locally without internet access.

Live Demonstration Sequence & User Interface

Step 1: Launch App

Double-click run.bat or execute python main.py. App initializes camera and local models.

Step 2: Interactive Overlay

High-contrast dark dashboard opens displaying real-time camera view and status panel.

Step 3: Press SPACE Bar

Captures frame, runs object detection + EasyOCR, composes description, and speaks out loud.

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Core Innovation & Project Differentiation

Zero-Cloud Local Pipeline

Complete perception and speech synthesis execute locally on-device with zero cloud telemetry.

Ultra-Low Storage Footprint

Sub-15 MB total model binaries ensure low memory overhead and rapid cold-start times.

Deterministic Response Merger

Synthesizes visual bounding boxes and OCR text into natural sentences with zero LLM hallucination.

QUALCOMM® SNAPDRAGON® AI LAB BUILD & PRESENT CHALLENGE

NETRA: "See it. Understand it. Hear it."

Empowering independence for low-vision users through Snapdragon-powered on-device AI.

Thank You! Submission package prepared for review.